

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

19th November 2020

Subject SA7 – Investment and Finance

Time allowed: 3 Hours 30 Minutes (14.30 - 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. *The answers are expected to be India Specific application for the syllabus and corresponding core reading. However, substantially the core reading material is still taken from material supplied by Actuarial Education Company which is meant for UK Fellowship examination. The core reading also contains some material which is India Specific, mostly the IRDA regulation. In view of this, it should be noted that focal point of answers is expected to be India Specific application. However, if application specific to any other country is quoted in the answer the candidate should answer the question with reference to Indian environment.*
3. *Mark allocations are shown in brackets.*

- Q. 1)** The National Government has been trying to increase the social benefits to its population, particularly the retired individual from private sector who does not have access to pension similar to the one given to Government employees. The Government is keen to introduce defined contribution pension scheme to be made available to all private sector employees and new employees joining the Government jobs. As part of regulations introduced by Government, the fund is being managed by a Board of Trustees. You are a consultant employed by the board for advising them on the Fund's investment strategy.

The Trustees have decided to adopt a Lifestage investment strategy. The Lifestage framework comprises three portfolios with different risk profiles. These portfolios have the following allocations:

	Growth Portfolio	Moderate Portfolio	Low Risk Portfolio
Equities	65%	45%	20%
Bonds	30%	45%	55%
Cash	5%	10%	25%

Members' funds are initially invested in the Growth Portfolio. At age 50 they are transferred to the Moderate Portfolio and at age 55 are transferred to the Low Risk Pre-retirement portfolio. Retirement age is 60. No further options are made available to members.

- i)** Discuss the limitations inherent in Lifestage investment strategy and recommend solutions to these. (10)

One of the Trustees believe that the fund should invest more widely and has proposed to invest fund in unquoted equities and commodities.

- ii)** Outline the advantages and disadvantages of investing in each of these new asset classes. (7)
- iii)** COVID 19 has hit country. The country had to impose several harsh lockdowns. As a result it is experiencing slow economic growth and relatively high unemployment, which is at the same time accompanied by rising prices (i.e. inflation).

Discuss the expected impact on the performances of the following assets.

- a)** Domestic Equities (4)

- b)** Commodity like Aluminum (3)

The Trustees are considering introducing a default annuity option for members retiring from the Fund.

- iv)** Briefly describe and list the advantages and disadvantages of the following options:
- a)** Fixed Life annuity (with or without fixed increases) (2)
- b)** Living annuity (2)
- c)** Inflation-linked annuity (2)
- d)** With-Profits annuity (1)

The Trustees have narrowed their options for the default annuity to either an Inflation-linked annuity or a With-Profits annuity provided by a reputable insurer.

- v) Recommend, with reasons, how the investment strategy for the Pre-Retirement Portfolio should be changed (if at all) in order to better protect the level of post-retirement income for each option. (5)
 - vi) A Trustee has advised that one of the Fund's investment managers, in a report-back, showed them a graph of the Indian Volatility Index (VIX). The trustee has asked for more information on the VIX.
 - a) Explain what the VIX represents and how the index should be interpreted. (3)
 - b) Describe the potential uses to an institutional investor of VIX as an indicator, as well as the derivative instruments based on it. (3)
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Q. 2) Two press articles recently caught your attention.

- i) First one talks about a non-professional investor who has made some investments. Describe the financial behaviors' exhibited by each of the scenario below.
 - a) He bought a banking stock 4 months back and it has lost 40% of its value. Market trends and consensus is that the share price will dip further. (2)
 - b) He bought a share in a pharma company after reading a report on the vaccine it has developed and awaiting approval for the trials. No major analyst is confident about the capability of the pharma company to develop this vaccine and the approvals to come through. (2)
 - c) He bought a stock in the travel portal Makemetrip which has risen in value from INR 100 to INR 200 per share over the last two years. Makemetrip was innovative in its offering but market becoming more competitive and it is likely that Makemetrip may lose customers. The investor continues to hold the stock as he believes that, based on past performance and his own research, the share price will double again over the next two years. (2)
 - d) He buys a bond with a 5% per annum guaranteed return. The alternative investment he could have made was in a bond with a 20% probability of a 0% return and an 80% chance of a return greater than 10% per annum. When discussing the bond investment with his wife, he states "we can either invest in a bond which will guarantee 5% each year or in a bond where there is a 20% chance we will not make any money". (2)
- ii) Second article that you read in the press recently has commented " The investment regulations are highly prescriptive"
 - a) List 2 example of investment regulation from the insurance industry which can be described as prescriptive. (2)
 - b) Give 2 advantages and 2 disadvantages of such prescriptive regulations? (4)

- c) Explain for a bank what the minimum reserve requirement is and how this differs from a risk-based capital requirement. (3)
- d) Give reasons why a bank regulator may prefer a risk-based capital requirement to a non risk-based capital measure with a higher level of coverage. (5)

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Q. 3) A life insurance company, Lead Life Pvt Co. (Lead Life), is considering buying Aam Bazaar Pvt, a network of life insurance web aggregator. Lead Life will be paying for Aam Bazaar with retained cash.

Aam Bazaar earns income through commission on the products it sells. The commission paid is high in the first year and smaller renewal commission is paid until expiry. Aam Bazaar has a significant amount of debt on its balance sheet and it has recently made a loss.

Aam Bazaar's Board of Directors rejected Lead Life's initial offer and Lead Life has announced publicly that it is considering whether to make a public offer for the shares of Aam Bazaar, that is a hostile bid for the company.

- i) Suggest reasons for the acquisition and why these reasons are likely to be borne out in practice. (12)
- ii) Describe defensive strategies that Aam Bazaar could employ to prevent the acquisition. (6)
- iii) Describe the factors that will affect the share prices of both Lead Life and Aam Bazaar during the bid period. (4)

Lead Life plans to securitise Aam Bazaar's commission income once the purchase is completed.

- iv) Describe how the securitization could be structured. (4)

Whilst the process was on, COVID 19 hit the country and Lead Life Co. decides to drop the bid. Though Aam Bazaar successfully defended the bid but the costs and management distraction of doing so have resulted in considerable financial strain.

- v) List possible management actions that may be taken as a result of Aam Bazaar being in financial distress. (5)
- vi) Outline the possible consequences of these actions. (5)

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